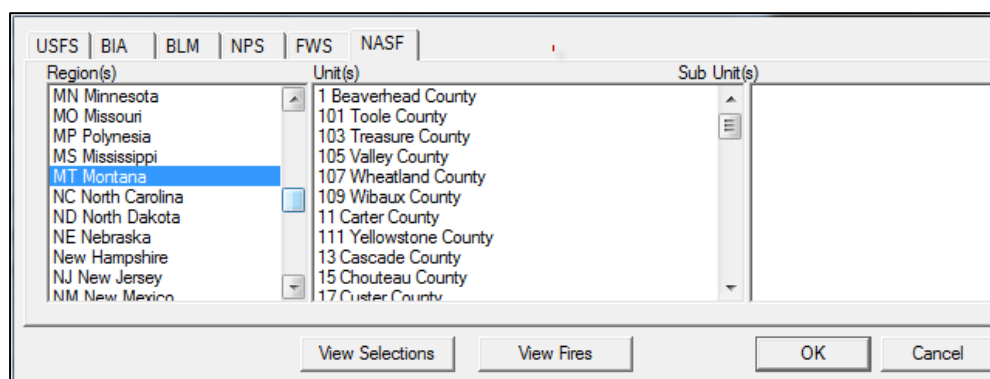


Obtaining and Importing NASF Fire Data into FireFamilyPlus Version 4.2

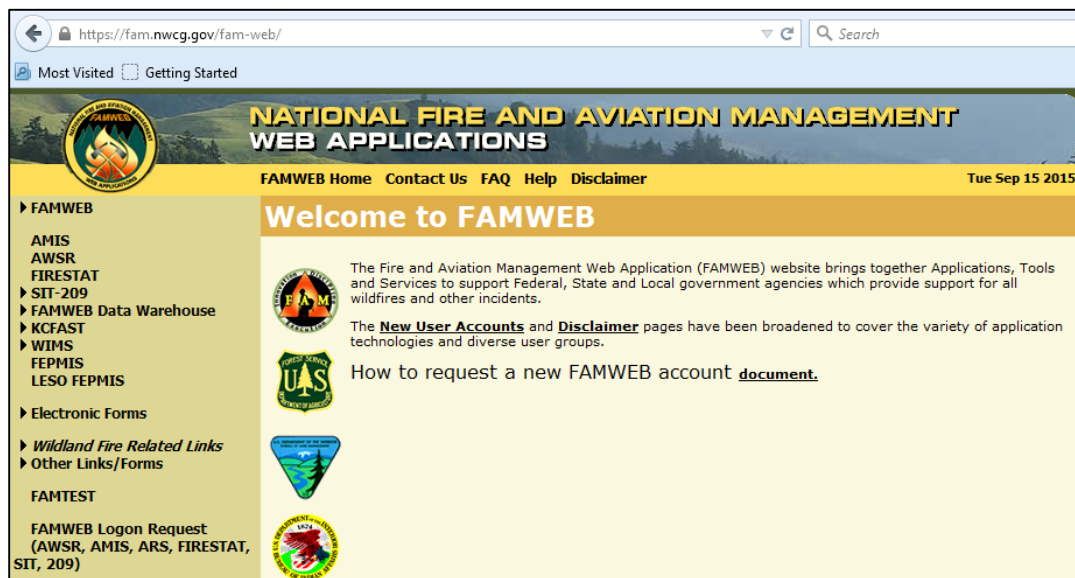
Version 4.2 of FireFamilyPlus has added the capability of importing state fire records from the FAMWEB Data Warehouse based on fire records from the National Association of State Foresters (NASF). Within FireFamilyPlus, the data are organized by State (region) and County (unit). There are no default Sub Unit structures. This tutorial describes how to access and download data to your computer and then import and use the data in FireFamilyPlus. Appendix A illustrates the degree of success importing records from 50 states over the period 2010 through 2014.



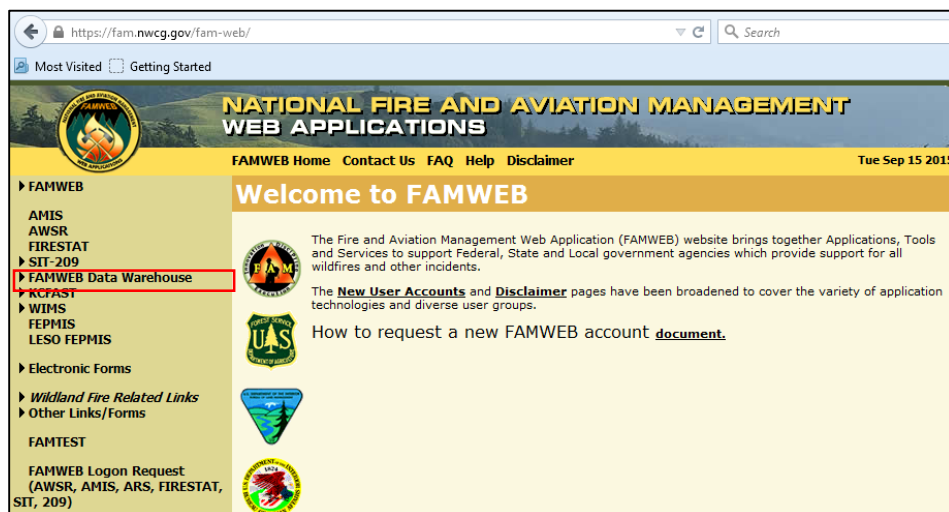
Accessing and extracting NASF fire reports from the FAMWEB Data Warehouse.....	2
Saving and reformatting NASF data files	7
Importing and working with NASF data in FireFamilyPlus.....	10
General fire summaries.....	12
Creating a fire summary graph	15
Appendix A.....	16

Accessing and extracting NASF fire reports from the FAMWEB Data Warehouse

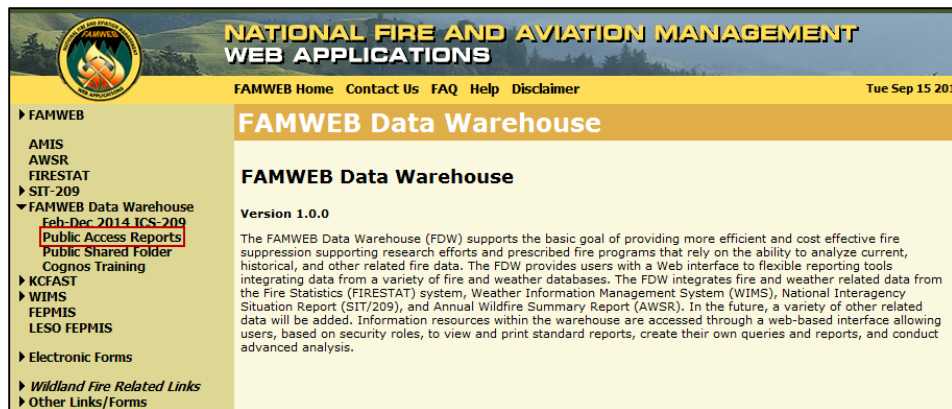
1. Access the [National Fire and Aviation Management Web Applications](https://fam.nwcg.gov/fam-web/) as site shown below.



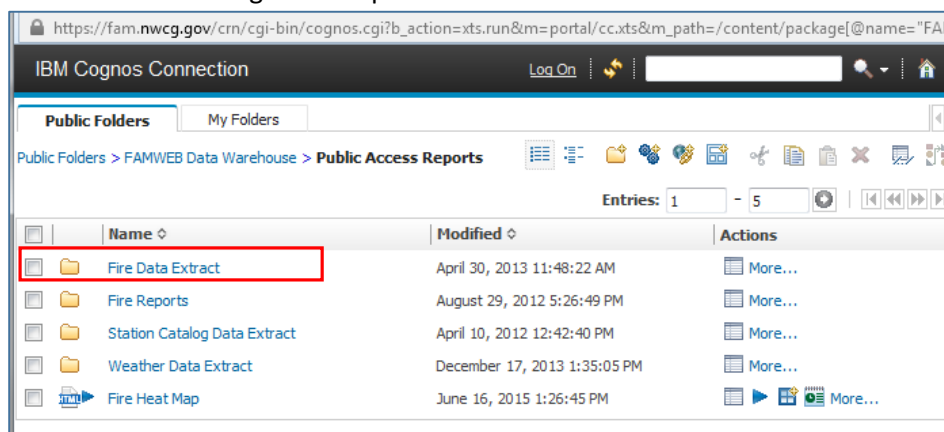
2. Select the **FAMWEB Data Warehouse** link on the left-hand menu bar as shown below to obtain state data.



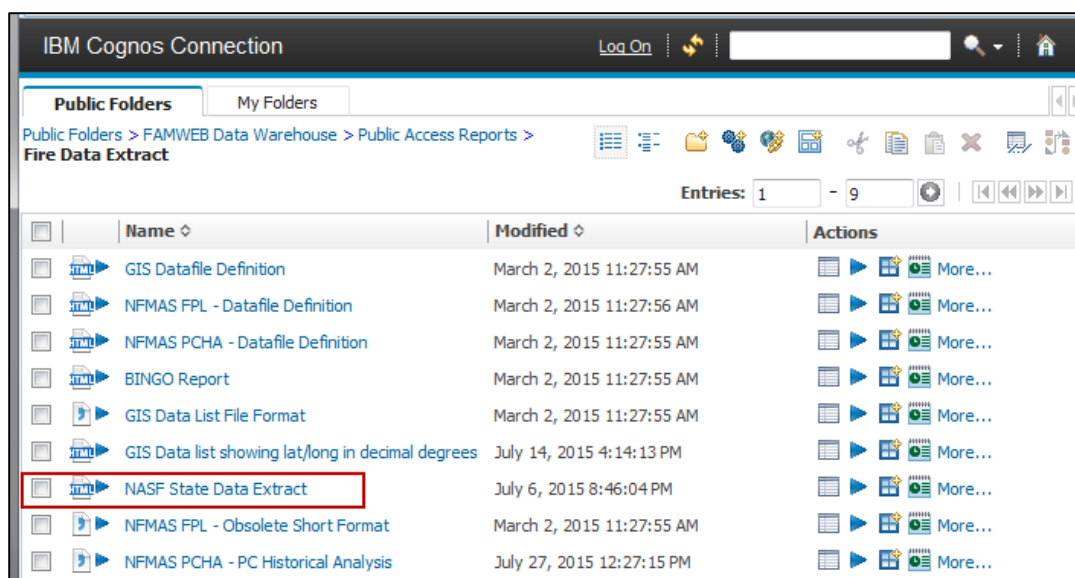
3. From the available selections, click on **Public Access Reports** as shown below.



4. The following screen opens. Click Fire Data Extract.



5. From the available selections, select NASF State Data Extract.



6. Use the drop-down menu to select a state or states of interest. (In this example Montana (MT) has been selected).

Prompt

Provide values for the report you are about to run

- * Indicates a required field.
- Points to missing information.

State

Provide a value:

LA

MA

MD

ME

MI

MN

MO

MS

MT

NC

ND

...

Select all Deselect all

7. Choose the fire year (or years) of interest using the drop-down menu provided to highlight your selections. In this example, the years 2000 through 2015 have been selected.

Fire Year

Provide a value:

1996

1997

1998

1999

2000

2001

2002

2003

2004

2005

2006

Select all Deselect all

8. When finished with your selection, click **OK** at the bottom of the page.

A **State Fire Occurrence Data Extract** report similar to the following example opens next. This report is an example of the data you are about to extract.

IBM Cognos Viewer - NASF State Data Extract

Log On | | | | About IBM.

Keep this version | | | | Add this report |

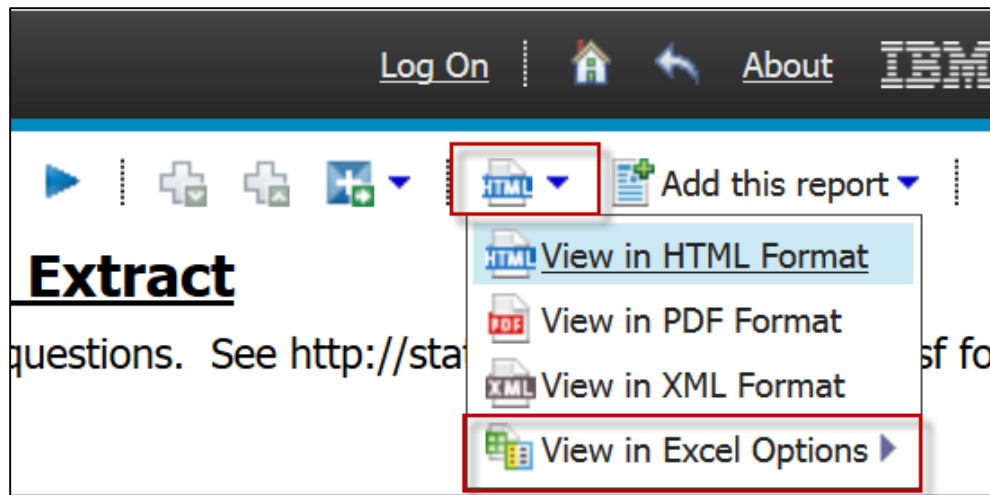
State Fire Occurrence Data Extract

This data is not official. Please contact the State Forestry Fire Program for official data and answers to questions. See http://stateforesters.org/about_nasf for more information.

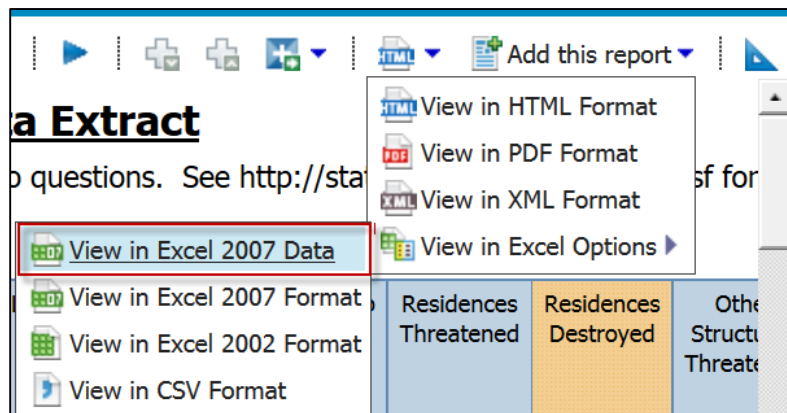
State: MT AND Fire Year: 2000, 2001, 2002, 2003, 2004, 2005, 2006, 2007, 2008, 2009, 2010, 2011, 2012, 2013, 2014, 2015

Local Incident ID	Fire Discovery Date	Incident Name	Fire Discovery Time	Fire Containment Date	Fire Containment Time	Fire Reporting Agency Unit ID	State	State FIPS	County	County FIPS	District	Latitude	Longitude	Statistical Cause Code	Ownership Code	Residences Threatened	Residences Destroyed	Other Structures Threatened	Other Structures Destroyed	Number Injuries	Number Fatalities	
000001	01/06/2000	MILLENNIUM	0015	01/07/2000	0000	MTCES	MT	30	Jefferson	043	CES	46.5061	-111.9275	4	F		0	0	0	0	0	0
000001	03/27/2000	FOLSOM	0000	03/28/2000	0000	MTSWS	MT	30	Missoula	063	SWS	46.7205	-114.1118	5	P		0	0	0	0	0	0
000001	04/16/2000	PEOPLES CREEK	0000	04/17/2000	0030	MTNWS	MT	30	Lincoln	053	NWS	48.3397	-115.3201	4	P		0	0	0	0	0	0
000001	06/29/2000	AIRPORT FIRE	0048	06/29/2000	0000	MTSOS	MT	30	Yellowstone	111	SOS	45.8221	-108.5618	9	P		0	0	0	0	0	0
000001	07/03/2000	WYANT COULEE	0000	07/08/2000	0000	MTEAS	MT	30	Rosebud	087	EAS	46.2015	-106.8932	1	P		0	0	0	0	0	0
000001	03/21/2001	BATAVIA	0000	03/21/2001	0000	MTNWS	MT	30	Flathead	029	NWS	48.1695	-114.4254	5	P		0	0	0	0	0	0
000001	03/22/2001	WEST CARLTON CREEK	0039			MTSWS	MT	30	Missoula	063	SWS	46.6732	-114.0908	5	P		0	0	0	0	0	0
000001	03/23/2001	BIRTHDAY "50"	0050	03/23/2001	0031	MTCES	MT	30	Jefferson	043	CES	46.2974	-112.1267	5	P		0	0	0	0	0	0
000001	05/05/2001	SVENSON RANCH	0000	05/10/2001	0000	MTSOS	MT	30	Stillwater	095	SOS	45.8359	-109.5351	5	P		0	0	0	0	0	0
000001	05/14/2001	TWIN COULEE	0000	05/20/2001	0000	MTNES	MT	30	Golden Valley	037	NES	46.4208	-108.8735	1	P		0	0	0	0	0	0

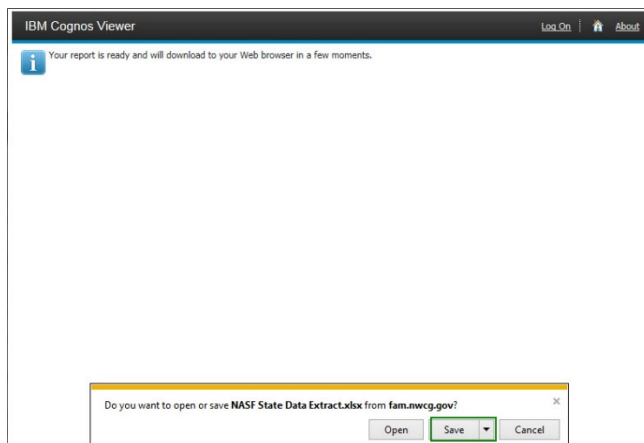
- Extract the data as a file to import into FireFamilyPlus. To accomplish this, select the drop-down arrow to the right of the **HTML** icon at the top right-hand side of the page as shown below. From the available selections, click on the last option, **View in Excel Options**.



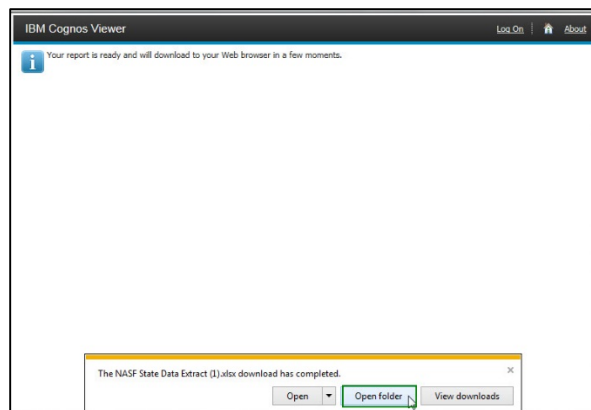
10. From the next menu selection, click on the first option **View in Excel 2007 Data**.



11. The following **Viewer** screen opens next. Click on the **Save** icon as shown below.



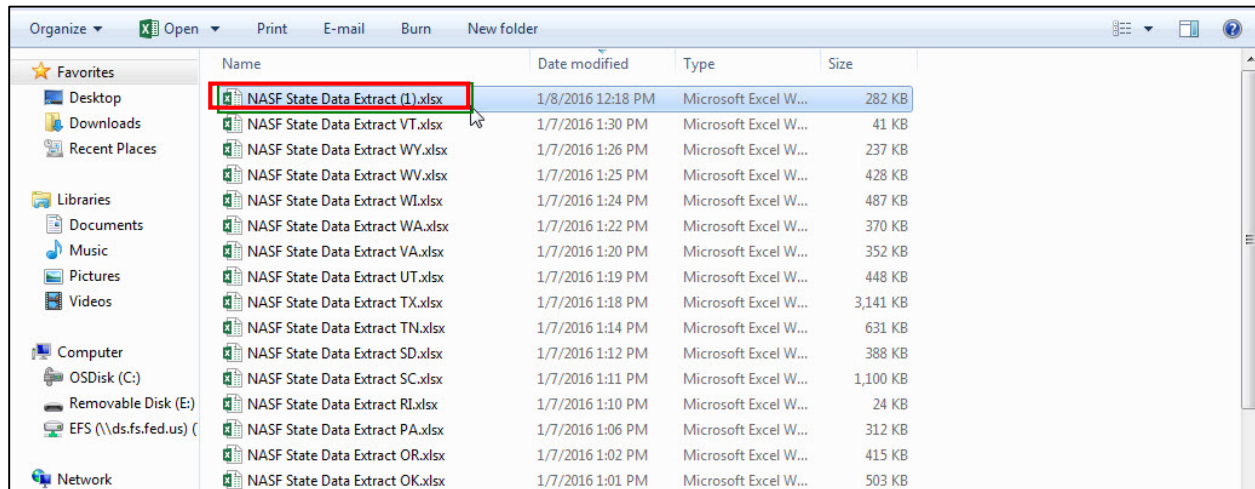
12. A notification box in the following dialog box which opens next indicates that the download has been completed. Click on the **Open folder** box.



Saving and reformatting NASF data files

FireFamilyPlus requires a comma delimited (csv) file but the Data Warehouse only exports Excel files. So you need to manipulate the Excel file and then save it as a CSV file.

13. The NASF State Data Extract has completed and the file is likely located in the **Downloads** folder. You will note the file is named “NASF State Data Extract” as. It is good practice to rename to reflect its content (i.e. State) as shown below



14. Double-click on the new **NASF State Data Extract.xlsx** file to open Microsoft Excel and view the file. The file opens as shown in the following example.

FileHomeInsertPage LayoutFormulasDataReviewViewAcrobat

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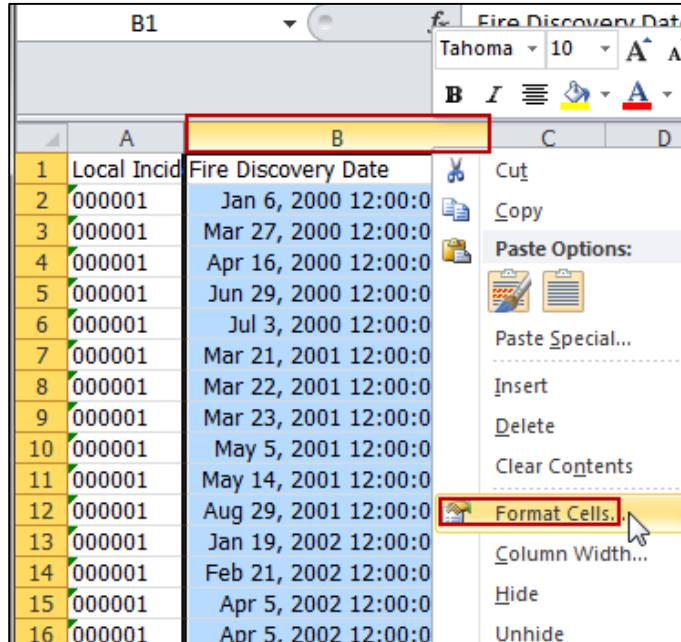
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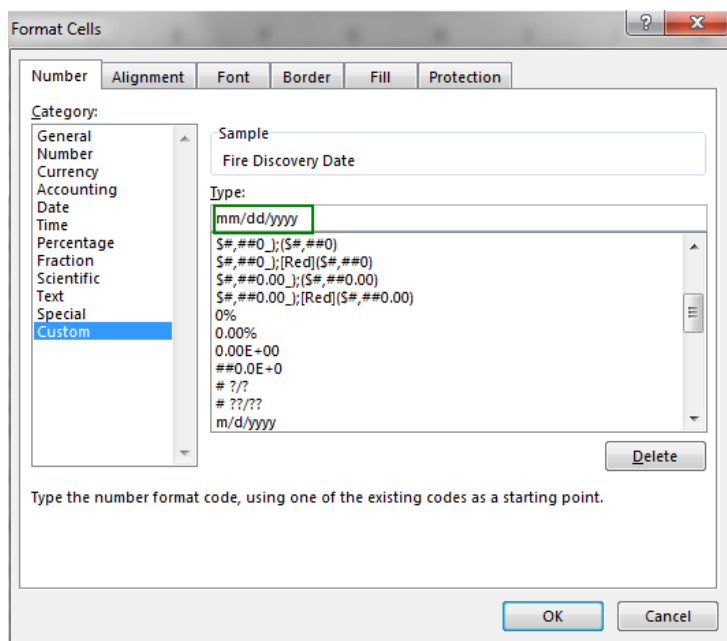
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15. Notice the date format in both columns “B” (Fire Discovery Date) shown in this example and “E” (Fire Containment Date). The default format is as follows: “Jan 6, 2000.” We will need to modify the date format **before** we can save the file with a “.CSV” extension. Right-click on column “B” and select “**Format Cells...**” as shown.



16. The **Format Cells** box opens as follows. On the **Number** tab, click **Custom** as the Category. Manually enter the format “**mm/dd/yyyy**” in the **Type** field. Click **OK**.



17. Repeat the same process for column E (Fire Containment Date). When finished, click **OK** again. In files from some states column E could not be reformatted. Since FFP does not need containment date, it is recommend that you delete all the values in this column (but keep the column heading). Do not delete the entire column.

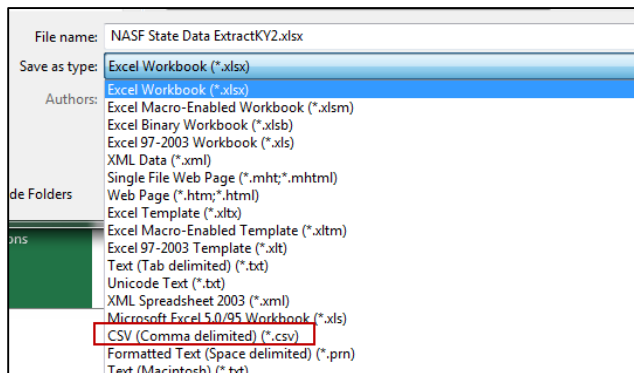
Hint: Once column B is correctly formatted and still selected, click format painter



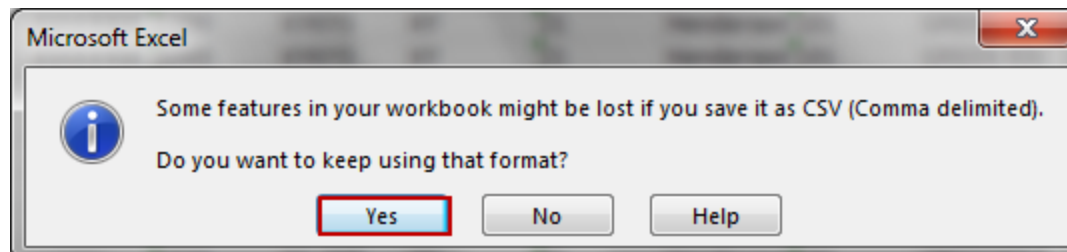
Format Painter

on the toolbar and click on column E to reformat Fire Containment Date.

18. In Excel, save the file to a folder of your choice using the drop-down list to specify a comma delimited (.csv) file (**File > Save As > CSV (comma delimited)**).



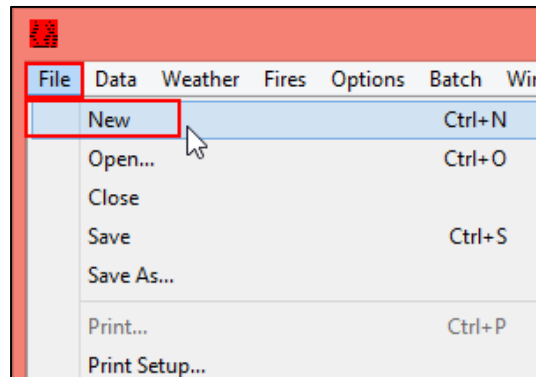
19. Select **Yes** if you see a message similar to the following example.



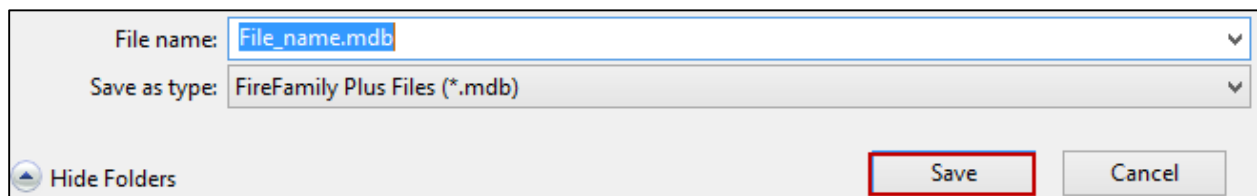
Importing and working with NASF data in FireFamilyPlus

The next step is to import your fire occurrence data into FireFamilyPlus. This example demonstrates creating a new database, but you can use an existing database (version 4.2 or higher).

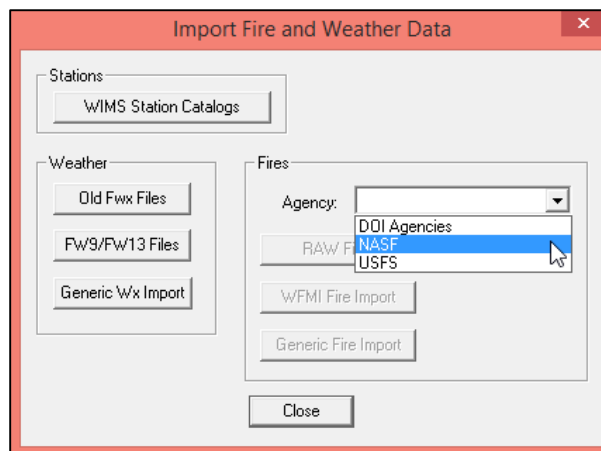
1. Open FireFamilyPlus and create a new database (Select **File > New** from the top menu). Or use an existing database (**File > Open**) if you prefer.



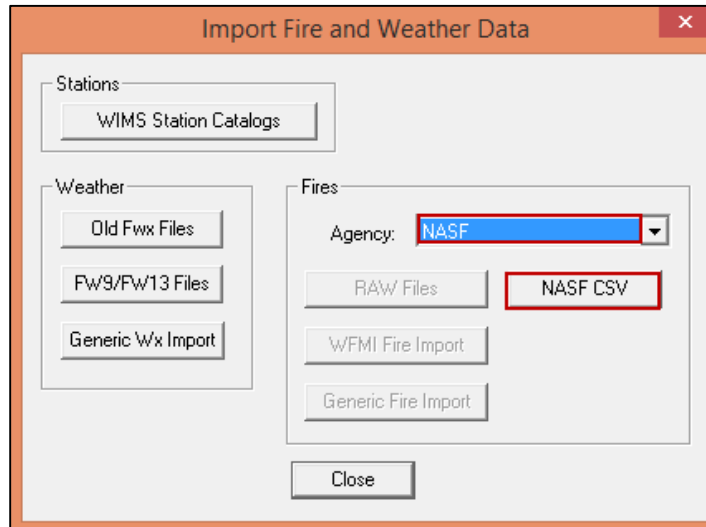
2. A new database opens. Name the database and specify a location to save it. Click **Save** after entering a descriptive name.



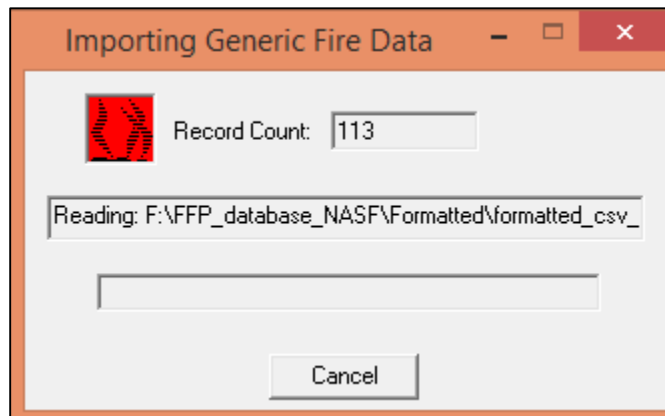
3. Select **Data > Import** on the top menu bar to open the **Import Fire and Weather Data** dialog box as shown below. Use the drop-down menu under **Fires, Agency:** to specify **NASF**.



4. After selecting **NASF**, a new icon, “**NASF CSV**” appears as an enabled option. Click on this icon and navigate to the NASF Fire file (.csv format) that you generated earlier.

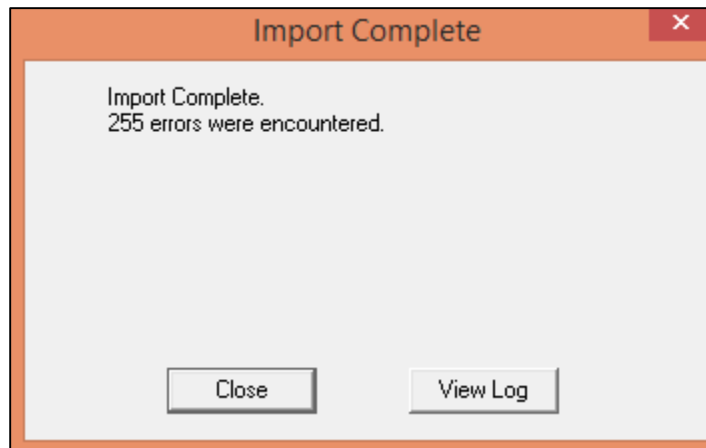


A record count in the **Importing Generic Fire Data** box indicates that the data is being processed.



5. When the import is finished, an **Import Complete** box opens, allowing you to view, print, or save an import error log. Click **View Log** for specific information on the types of errors encountered.

Note: Problems have been observed in NASF exports for some states. FireFamilyPlus makes no guarantee as to the quality of the State Data, particularly with respect to county and cause class. Appendix A provides a brief summary of the general type and scope of problems found for each state for 2010-14.

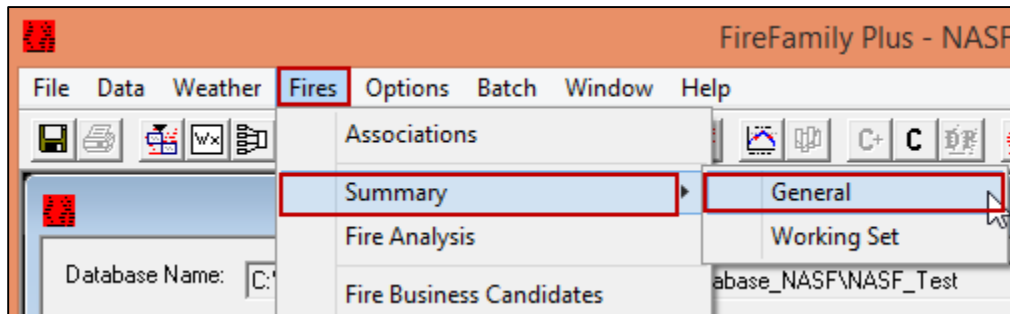


6. Click **Close** again to return the Working Set screen.

General fire summaries

FireFamilyPlus allows you to perform general fire summaries on fires that are not associated with a typical working set (station, years, and annual filter). The query allows you to select fire associations and years.

1. To perform a general fire summary with the NASF files you have imported click on **Fires** on the main menu. Select **Summary** and **General** from the drop-down list as shown. (**Fires > Summary > General**).



2. A **Select Fires and Years for General Summary** dialog box opens as follows. Click on the **NASF** tab at the top of the screen.

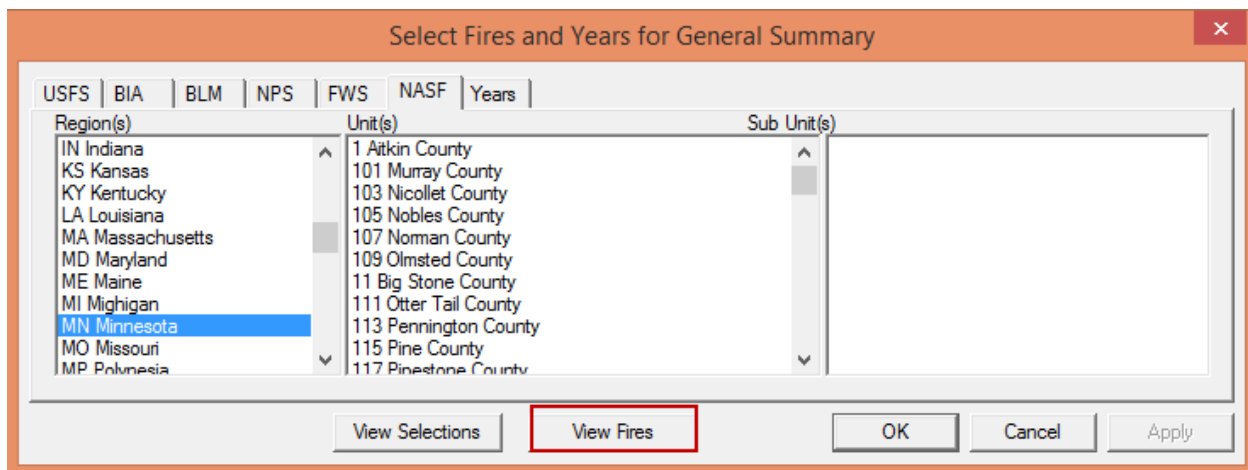
Region(s)	Unit(s)	Sub Unit(s)
01 Northern Region (R1)		
02 Rocky Mountain Region (R2)		
03 Southwest Region (R3)		
04 Intermountain Region (R4)		
05 Pacific Southwest Region (R5)		
06 Pacific Northwest Region (R6)		
08 Southern Region (R8)		
09 Eastern Region (R9)		
10 Alaska Region (R10)		

- Initially no regions or units are highlighted or selected. Specify the state or states of interest by highlighting your selections in the **Region(s)** column at the far left. To further refine your selection, you can also highlight **Unit(s)** to specify counties of interest. (MN, Minnesota is shown in the following example).

Region(s)	Unit(s)	Sub Unit(s)
MA Massachusetts	101 Murray County	
MD Maryland	103 Nicollet County	
ME Maine	105 Nobles County	
MI Michigan	107 Norman County	
MN Minnesota	109 Olmsted County	
MO Missouri	11 Big Stone County	
MP Polynesia	111 Otter Tail County	
MS Mississippi	113 Pennington County	
MT Montana	115 Pine County	
NC North Carolina	117 Pipestone County	
ND North Dakota	119 Polk County	

Note: The **Years** tab can be used to filter out unwanted records or to narrow your selection by specifying fires within only certain years. It is good practice to **View Selections** to ensure only the desired fires are displayed.

- To view a list of fires in the general fires summary, click on the **View Fires** icon.



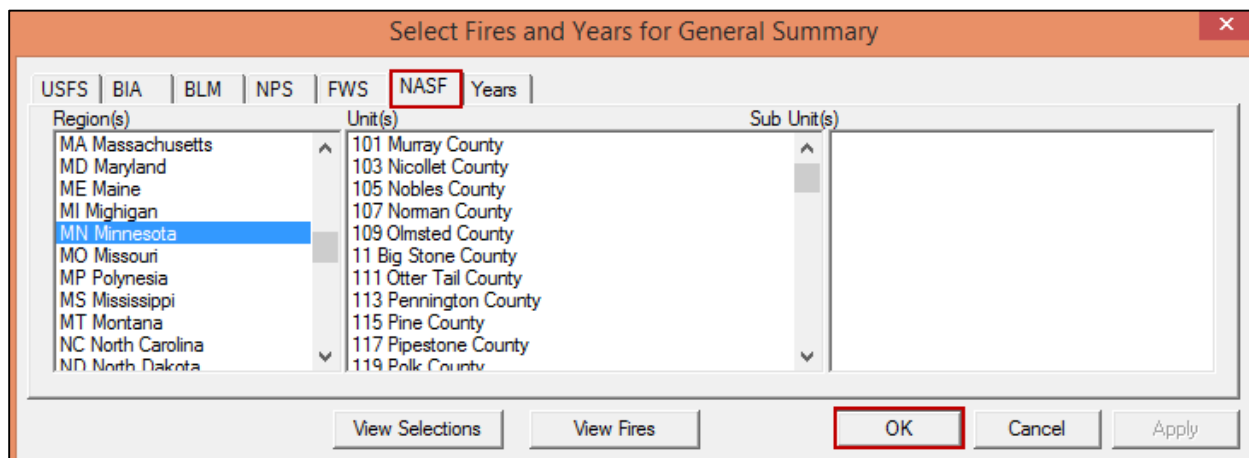
- A **Fires Table** similar to the following examples opens, displaying all of the selected fires.

	Discovery	Acres	Cause	Fire Name	Fire Num	Lat. (Deg)	(min)	(sec)	Lon. (Deg)	(min)	(sec)	Agency	Region	Unit
1	03/14/10	1.0	5 Debris Bur		2010161001	46	44	3	-94	57	44	NASF	MN Minnesota	159 Wadena Co
2	03/17/10	2.0	7 Arson		2010244001	46	0	37	-92	34	38	NASF	MN Minnesota	115 Pine Count
3	03/17/10	1.0	7 Arson		2010312001	45	57	12	-94	6	6	NASF	MN Minnesota	97 Morrison Coi
4	03/17/10	7.0	9 Miscellaneous		2010111001	47	20	30	-95	6	2	NASF	MN Minnesota	57 Hubbard Coi
5	03/17/10	1.0	5 Debris Bur		2010161002	46	25	16	-94	50	40	NASF	MN Minnesota	159 Wadena Co
6	03/17/10	3.0	7 Arson		2010312029	45	52	8	-93	53	48	NASF	MN Minnesota	97 Morrison Coi
7	03/18/10	1.0	5 Debris Bur		2010232001	46	38	44	-93	48	7	NASF	MN Minnesota	35 Crow Wing C
8	03/18/10	1.0	5 Debris Bur		2010312003	46	2	23	-94	14	12	NASF	MN Minnesota	97 Morrison Coi
9	03/18/10	1.0	5 Debris Bur		2010351001	45	28	39	-93	49	22	NASF	MN Minnesota	141 Sherburne
10	03/18/10	2.0	2 Equipmen		2010312002	45	41	37	-93	48	38	NASF	MN Minnesota	9 Benton Count
11	03/18/10	1.0	5 Debris Bur		2010142002	46	20	19	-94	21	4	NASF	MN Minnesota	21 Cass County
12	03/18/10	1.0	6 Railroad		2010351002	45	26	27	-93	15	31	NASF	MN Minnesota	59 Isanti Count
13	03/18/10	0.0	5 Debris Bur		2010142001	46	50	50	-94	22	35	NASF	MN Minnesota	21 Cass County
14	03/18/10	1.0	5 Debris Bur		2010251001	46	32	8	-92	32	34	NASF	MN Minnesota	17 Carlton Cou
15	03/18/10	9.0	5 Debris Bur		2010163009	46	38	59	-95	14	21	NASF	MN Minnesota	111 Otter Tail C
16	03/18/10	0.0	9 Miscellaneous		2010244002	46	0	36	-92	40	14	NASF	MN Minnesota	115 Pine Count
17	03/18/10	0.0	7 Arson		2010234001	47	29	21	-92	53	22	NASF	MN Minnesota	137 St. Louis Co
18	03/18/10	41.0	5 Debris Bur		2010232003	46	53	52	-93	40	30	NASF	MN Minnesota	1 Aitkin County
19	03/19/10	4.0	2 Equipmen		2010111002	47	27	42	-95	6	3	NASF	MN Minnesota	7 Beltrami Cour
20	03/19/10	0.0	5 Debris Bur		2010232002	46	21	37	-94	17	2	NASF	MN Minnesota	35 Crow Wing C
21	03/19/10	5.0	7 Arson		2010234002	47	22	22	-93	9	1	NASF	MN Minnesota	61 Itasca Coun

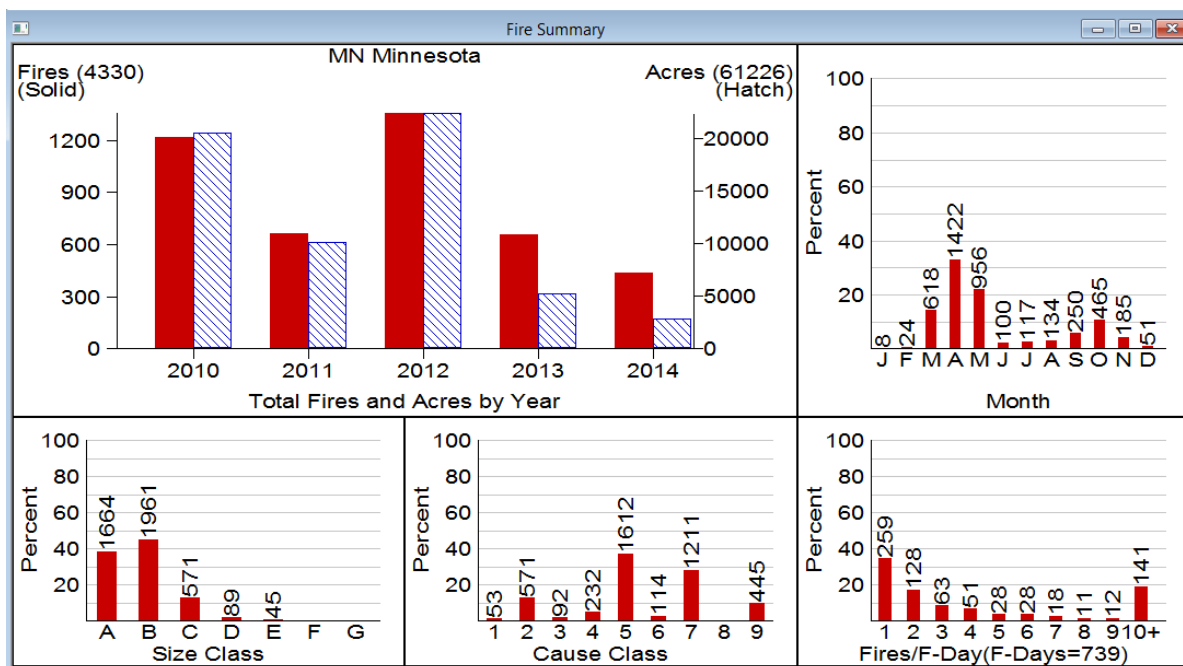
- The icons in this dialog box can be used to Export records (including shapefiles), Delete selected fire records from the table, or to save or print the entire table. (For additional information on how to carry out these functions, consult the FireFamilyPlus User's Guide or Help utility). **Close** the box to return to the Working Set.

Creating a fire summary graph

1. To generate a fire summary graph, click **OK** to graph the options you selected earlier.



2. A **Fire Summary** screen similar to the following example opens. The large graph at the upper left shows total number of fires and acres by year, the graph at the top right displays fires by month, and graphs in the lower portion of the screen show fires by size class, cause class, and fire days by number and percent. Graphs can be printed (**File > Print**), or saved as a number of possible image files (**File > Save As**). Options include viewing class definitions (**Options > View Class Definitions**) or viewing or printing graph data (**Options > Graph Data**).



Appendix A. Results of importing NASF fire records for all 50 states, 2010-2015 into FireFamilyPlus.

Records for 2010-2015 were extracted in February 2016 for each of the 50 states. Below are the results of importing the NASF files into FireFamilyPlus Version 4.2. No records for 2015 were returned.			
State	N Records	% Imported	Error Causes
AK	2,894	0	1
AL	16,516	25	1, 2
AR	9,647	52	1, 2
AZ	9,874	3	1
CA	30,591	61	2, 5
CO	23,129	95	2
CT	1,682	60	1, 5
DE	69	83	1
FL	15,081	92	1, 2
GA	26,554	99	1
HI	39	97	3
IA	2,300	61	1, 4
ID	1,181	58	1
IL	1,059	96	1, 2
IN	3,764	99	4, 5
KS	28,606	94	2
KY	6,443	99	1
LA	8,027	83	2, 4
MA	6,803	49	1, 2
MD	696	69	1, 2
ME	2,286	74	1
MI	5,117	84	2, 5
MN	4,585	94	2
MO	16,872	99	2
MS	9,556	57	1, 2
MT	1,523	94	2
NC	20,464	76	1, 2
ND	964	84	1, 2
NE	4,524	0	1
NH	930	68	1, 2
NJ	5,081	99	2
NM	3,095	91	2
NV	1,105	95	1, 2
NY	26,861	99	2
OH	2,244	63	1, 5
OK	5,354	99	1
OR	3,975	95	2, 5
PA	2,997	99	1
RI	187	50	1, 2, 4
SC	11,603	95	2
SD	3,956	98	1, 2, 5
TN	6,209	91	2
TX	31,192	93	1, 2, 4
UT	4,243	94	2
VA	3,870	99	1, 5
VT	400	83	1, 2, 5
WA	3,526	64	1, 2
WI	4,712	84	2
WV	3,926	83	1, 2
WY	2,358	76	1, 2

ERROR CODES	
1	Invalid or missing FIPS County Codes
2	Invalid or missing Statistical Cause Code (not USFS 1-9)
3	Missing Fire Number
4	Other
5	Column E cannot be reformatted